

Do **not** write solutions on this page.

11. [Maximum mark: 19]

The plane Π_1 has equation $2x + 6y - 2z = 5$.

- (a) Verify that the point $A\left(2, \frac{1}{2}, 1\right)$ lies on the plane Π_1 . [1]

The plane Π_2 is given by $(k^2 - 6)x + (2k + 3)y + pz = q$, where $p, q, k \in \mathbb{R}$ and $p \neq 0$.

- (b) In the case where $p = -6$, Π_2 is perpendicular to Π_1 and A lies on Π_2 . Find the value of k and the value of q . [5]

For parts (c), (d) and (e) it is now given that Π_2 is parallel to Π_1 with $k = 3$.

- (c) Determine the value of p . [2]

It is also given that $q = -\frac{51}{2}$.

The line through A that is perpendicular to Π_1 meets Π_2 at the point B .

- (d) (i) Find the coordinates of B .
(ii) Hence, show that the perpendicular distance between Π_1 and Π_2 is $\sqrt{11}$. [7]
- (e) Find the equation of a third parallel plane Π_3 which is also a perpendicular distance of $\sqrt{11}$ from Π_1 . [4]

